

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]

Tap amplitudes (pixels): [129.65338406690356, 129.5414991421668, 115.83177456984762, 110.01818031580054, 116.84605256490268, 107.33592129385204, 95.01578816175763, 116.76472069936193, 116.8289347721702, 114.12712210513327, 116.81181447096864]



Based on the extracted features provided, here is the analysis according to the MDS-UPDRS Item 3.4 (Finger Tapping) criteria:

1. Rhythm and Speed (Periods):

- The tap periods range from 92 to 123 frames, with a mean of approximately 103 frames.
- The rhythm shows some variation (standard deviation ~10.6 frames) and a trend of **hastening** (periods decreasing from 123 to 92 over the first half) rather than slowing.
- There are no clear "interruptions" or "halts" (no periods > 3-5x the mean). The rhythm is relatively continuous.

2. Amplitude (Pixels):

- **Tap 1 & 2:** High amplitude (~129 pixels).
- **Tap 3:** Significant drop to ~115 pixels (approx. 11% decrease).
- **Tap 4-11:** Amplitudes fluctuate mostly between 110 and 116 pixels, with a noticeable dip at Tap 7 (~95 pixels).
- **Trend:** The subject demonstrates a "sequence effect" where the amplitude drops after the initial successful attempts. Specifically, the high amplitude is lost after

the 2nd tap, and the subject settles into a smaller, faster rhythm (hastening).

Scoring Determination:

- **Score 0 (Normal):** Incorrect. There is a clear drop in amplitude and change in rhythm (hastening).
- **Score 1 (Slight):** Incorrect. The amplitude decrement occurs early in the sequence, not "near the end".
- **Score 2 (Mild):** Incorrect. The criteria for Mild require the amplitude to decrement "midway" (around tap 5). Here, the decrement begins at Tap 3.
- **Score 3 (Moderate):** Correct. The criteria state: *"The amplitude decrements starting after the 1st tap."* In this dataset, the decrement begins clearly at Tap 3 (immediately following the first two taps), which aligns with the "early" onset defined in this category. The combination of early amplitude loss and hastening rhythm fits the profile of moderate impairment.
- **Score 4 (Severe):** Incorrect. The subject completes the task without freezing.

Score: 3